Combinatorics 9

Siddharth K

November 2025

Problem 1

We have that

$$\left(\sum_{n=0}^{\infty}\frac{x^n}{n!}\right)\left(\sum_{m=0}^{\infty}\frac{y^m}{m!}\right)=\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\frac{x^n y^m}{n!\,m!}.$$

Relabeling indices as $n := n+m$ and $k := n$, the sum equals

$$\sum_{n=0}^{\infty}\sum_{k=0}^{n}\frac{x^k y^{n-k}}{k!(n-k)!}=\sum_{n=0}^{\infty}\frac{1}{n!}\sum_{k=0}^{n}\frac{n!}{k!(n-k)!}x^k y^{n-k}=\sum_{n=0}^{\infty}\frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k}$$

$$=\sum_{n=0}^{\infty}\frac{1}{n!}(x+y)^n,$$

as desired.

Now, assume we know that $e^x e^y = e^{x+y}$. Expanding each side as a power series, the LHS becomes

$$\left(\sum_{n=0}^{\infty}\frac{x^n}{n!}\right)\left(\sum_{m=0}^{\infty}\frac{y^m}{m!}\right)=\sum_{n=0}^{\infty}\sum_{k=0}^{n}\frac{x^k y^{n-k}}{k!(n-k)!},$$

and the RHS becomes

$$\sum_{n=0}^{\infty}\frac{(x+y)^n}{n!}.$$

Since the two sides are equal, the power series must agree term by term. So fix $n \geq 0$. Equating the coefficient of $x^k y^{n-k}$ on both sides gives us

$$\frac{1}{k!(n-k)!}=\frac{1}{n!}(\text{coefficient of } x^k y^{n-k} \text{ in } (x+y)^n).$$

Therefore, the coefficient is

$$\frac{n!}{k!(n-k)!}=\binom{n}{k}.$$

So for every $n \geq 0$,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k},$$

which is the binomial theorem.

Problem 2

We have that

$$A'(x) = \sum_{n\geq 0} a_n \frac{x^{n-1}}{(n-1)!} = \sum_{n\geq -1} a_{n+1} \frac{x^n}{n!}$$

$$= \sum_{n\geq -1} a_{n+1} \frac{x^n}{n!} = \sum_{n\geq 0} a_{n+1} \frac{x^n}{n!} - a_0 \frac{x^{-1}}{(-1)!} = \sum_{n\geq 0} a_{n+1} \frac{x^n}{n!},$$

so

$$\sum_{n\geq 0} a_{n+1} \frac{x^n}{n!} = A'(x).$$

Problem 3

$$\zeta(s)^k = \sum_{n=1}^{\infty} \frac{d_k(n)}{n^s}$$

where $d_k(n)$ is the number of ordered k-tuples with elements multiplying to n. We can prove this by induction.

If $k = 1$ then $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, which always has coefficient 1, which counts the number of single numbers whose product is n (which is (n)). Now assume for some $k \geq 1$ that

$$\zeta(s)^k = \sum_{n=1}^{\infty} \frac{d_k(n)}{n^s}.$$

Then

$$\zeta(s)^{k+1} = \zeta(s)^k \zeta(s) = \left(\sum_{a=1}^{\infty} \frac{d_k(a)}{a^s} \right) \left(\sum_{b=1}^{\infty} \frac{1}{b^s} \right)$$

$$= \sum_{n=1}^{\infty} \left(\sum_{a|n} d_k(a) \right) \frac{1}{n^s}.$$

So the coefficient of $1/n^s$ is

$$\sum_{a|n} d_k(a).$$

This is equal to $d_{k+1}(n)$ because it sets a as one of the $k+1$ divisors and counts the k remaining ones using $d_k(a)$, and finally sums over a.

Problem 5

Define

$$B(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s}.$$

We have that $B(s) = A(s)\zeta(s)$, since

$$A(s)\zeta(s) = \left(\sum_{n=1}^{\infty} \frac{a_n}{n^s}\right)\left(\sum_{m=1}^{\infty} \frac{1}{m^s}\right) = \sum_{n=1}^{\infty} \left(\sum_{d|m} a_d\right) \frac{1}{n^s}$$

$$= \sum_{n=1}^{\infty} \frac{b_n}{n^s} = B(s).$$

Problem 6

Define

$$F(x) = \sum_{n=0}^{\infty} F_n \frac{x^n}{n!}.$$

Then we have that

$$F'(x) = \sum_{n=0}^{\infty} F_{n+1} \frac{x^n}{n!}, \quad F''(x) = \sum_{n=0}^{\infty} F_{n+2} \frac{x^n}{n!}.$$

Using the Fibonacci recurrence,

$$\begin{aligned} F_{n+2} &= F_{n+1} + F_n, \\ \sum_{n=0}^{\infty} F_{n+2} \frac{x^n}{n!} &= \sum_{n=0}^{\infty} F_{n+1} \frac{x^n}{n!} + \sum_{n=0}^{\infty} F_n \frac{x^n}{n!}, \\ F''(x) &= F'(x) + F(x), \\ F''(x) - F'(x) - F(x) &= 0. \end{aligned}$$

Solving the ODE with initial conditions $F(0) = F_0 = 0$ and $F'(0) = F_1 = 1$, we get

$$\alpha,\, \beta = \frac{1 \pm \sqrt{5}}{2},$$

$$F(x) = C_1 e^{\alpha x} + C_2 e^{\beta x}.$$

From $F(0) = 0$, we get that $C_2 = -C_1$. From $F'(0) = 1$, we have that

$$C_1\alpha + C_2\beta = 1,$$

$$C_1 = \frac{1}{\alpha - \beta} = \frac{1}{\sqrt{5}}, \quad C_2 = -\frac{1}{\sqrt{5}}.$$

So

$$F(x) = \frac{1}{\sqrt{5}}(e^{\alpha x} - e^{\beta x}).$$

Expanding this as a power series, we get

$$\sum_{n=0}^{\infty} F_n \frac{x^n}{n!} = \frac{1}{\sqrt{5}} \left(\sum_{n=0}^{\infty} \alpha^n \frac{x^n}{n!} - \sum_{n=0}^{\infty} \beta^n \frac{x^n}{n!} \right) = \sum_{n=0}^{\infty} \frac{1}{\sqrt{5}} (\alpha^n - \beta^n) \frac{x^n}{n!},$$

$$F_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n).$$

Problem 8

A permutation can be expressed as a set of cycles, so a derangement is a set of cycles of length ≥ 2. The number of cycles of length k is $(k-1)!$, so the term in an exponential generating function would be $(k-1)!\frac{x^k}{k!} = \frac{x^k}{k}$. So the exponential generating function encoding the number of cycles of length ≥ 2 is

$$\sum_{k=2}^{\infty} \frac{x^k}{k} = \left(\sum_{k=1}^{\infty} \frac{x^k}{k} \right) - x = -\log(1-x) - x.$$

A derangement is a set of such cycles, so its exponential generating function is

$$D(x) = e^{-\log(1-x)-x} = e^{-x}(1-x)^{-1} = \frac{e^{-x}}{1-x}.$$

To extract the number of derangements, D_n, we can unsum each factor:

$$e^{-x} = \sum_{j=0}^{\infty} \frac{(-1)^j x^j}{j!}, \quad \frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

$$\frac{e^{-x}}{1-x} = \left(\sum_{j=0}^{\infty} \frac{(-1)^j x^j}{j!} \right) \left(\sum_{k=0}^{\infty} x^k \right) = \sum_{n=0}^{\infty} \left(\sum_{j=0}^{n} \frac{(-1)^j}{j!} \right) x^n.$$

But we also have that

$$D(x) = \sum_{n=0}^{\infty} D_n \frac{x^n}{n!},$$

so

$$\frac{D_n}{n!} = \sum_{j=0}^{n} \frac{(-1)^j}{j!},$$

$$D_n = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}.$$

Problem 10

We have that

$$E(tx) = \sum_{n=0}^{\infty} a_n \frac{t^n x^n}{n!}.$$

Substituting into the integral, we get

$$\int_0^{\infty} e^{-x} \left(\sum_{n=0}^{\infty} a_n \frac{t^n x^n}{n!} \right) = \sum_{n=0}^{\infty} \left(a_n \frac{t^n}{n!} \int_0^{\infty} e^{-x} x^n dx \right).$$

By repeating integration by parts, we have that

$$\int_0^{\infty} e^{-x} x^n dx = n!,$$

so the sum becomes

$$\sum_{n=0}^{\infty} a_n t^n = A(t).$$

Problem 16

First, we find a recurrence relation for a_n. Consider an alternating permutation π of $[n+1]$. The value $n+1$ must be at an even position because it's the greatest value. If we split the permutation at the position of $n+1$, we have

$$\pi = (\text{left block})\,(n+1)\,(\text{right block}).$$

The left and right blocks are each an alternating permutation starting with $<$, because $n+1$ is greater than the first element in the right block.

To form π, we can choose a subset of $[n]$ of size k in $\binom{n}{k}$ ways, form one block in a_k ways, and then form the other block in a_{n-k} ways. For each pair of blocks, we get two alternating permutations of $[n+1]$ since each one can be a left block or right block. So we have that

$$a_{n+1} = \frac{1}{2} \sum_{k=0}^{n} \binom{n}{k} a_k a_{n-k},$$

for $n \geq 1$ with $a_0 = 1$ and $a_1 = 1$.

Now, we have that

$$A'(x) = \sum_{n=0}^{\infty} a_{n+1} \frac{x^n}{n!} = a_1 + \sum_{n=1}^{\infty} a_{n+1} \frac{x^n}{n!}$$

$$= a_1 + \frac{1}{2} \sum_{n=1}^{\infty} \sum_{k=0}^{n} \binom{n}{k} a_k a_{n-k} \frac{x^n}{n!} = a_1 + \frac{1}{2} \left(\sum_{n=0}^{\infty} \sum_{k=0}^{n} \binom{n}{k} a_k a_{n-k} \frac{x^n}{n!} - \sum_{k=0}^{0} \binom{0}{k} a_0 a_1 \right)$$

$$= 1 + \frac{1}{2}\left(\sum_{k=0}^{\infty} a_k \frac{x^k}{k!}\right)\left(\sum_{m=0}^{\infty} a_m \frac{x^m}{m!}\right) - \frac{1}{2} = \frac{1}{2}(A(x)^2 + 1),$$

$$A'(x) = \frac{1}{2}(A(x)^2 + 1).$$

Solving this differential equation, we get

$$\frac{2}{A^2+1}\frac{dA}{dx} = 1, \quad \int \frac{2}{A^2+1}\frac{dA}{dx}dx = \int dx,$$

$$2\arctan A = x + C.$$

From $A(0) = a_0 = 1$,

$$2\arctan 1 = 0 + C, \quad C = \frac{\pi}{2}.$$

So

$$A(x) = \tan\left(\frac{x}{2} + \frac{\pi}{4}\right),$$

which, according to Desmos, appears to be equal to $\sec x + \tan x$.